21. Yoshida T, Nagamine T, Kobayashi T, et al. Impairment of the inferior alveolar nerve after sagittal split osteotomy. *J Craniomaxillofac Surg* 1989;17:271–277
22. McLeod NMH, Bowe DC. Nerve injury associated with orthognathic surgery. Part 2: inferior alveolar nerve. *Br J Oral Maxillofac Surg* 2016;54:366–371
23. Reyneke JP, Tsakiris P, Becker P. Age as a factor in the complication rate after removal of unerupted/impacted third molars at the time of mandibular sagittal split osteotomy. *J Oral Maxillofac Surg* 2002;60:654–659
24. Hanzelka T, Foltán R, Pavlíková G, et al. The role of intraoperative positioning of the inferior alveolar nerve on postoperative paresthesia after bilateral sagittal split osteotomy of the mandible: prospective clinical study. *Int J Oral Maxillofac Surg* 2011;40:901–906
25. Westermark A, Englesson L, Bongenhielm U. Neurosensory function after sagittal split osteotomy of the mandible: a comparison between subjective evaluation and objective assessment. *Int J Adult Orthodont Orthognat Surg* 1999;14:268

An Accurate Volumetric Analysis Method for Evaluating Outcomes of Alveolar Cleft Reconstruction

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Abstract: An accurate volumetric analysis method for evaluating the outcomes of different types of alveolar cleft reconstruction is essential because it can help determine which graft material is more effective, confirm favorable times for alveolar bone grafting, and improve surgical techniques. This study aimed to introduce a novel method of precisely calculating the bone formation ratio using computer-aided engineering after surgery. A patient with a unilateral alveolar cleft who was treated with anterior iliac crest bone grafting was enrolled in this study. Helical computed tomography scans were performed preoperatively and 12 months postoperatively. The Digital Imaging and Communications in Medicine (DICOM) data were reconstructed as three-dimensional images and saved in the STL format by using Mimics software. STL data were processed by Geomagic Wrap 2017, using the Boolean operation, the newly formed bone of the alveolar was segmented by identifying the differences between the preoperative and the postoperative three-dimensional images. For this patient, the mean volume of the newly formed bone was 0.387 cm^3, the morphology was clear, the bone formation ratio was 41.4%, the mean time required for calculating the newly formed bone volume was 23 minutes, and the bone survival ratio was 38.7%. This method is a clinically practical, accurately measurement and time-saving method to evaluate the outcome of alveolar cleft reconstruction. Both the volumetric assessment and morphological analysis of the newly formed bone could be determined in a precise manner.

Key Words: Alveolar cleft, computer-aided engineering, volumetric analysis

The alveolar cleft is a common maxillary bone defect resulting from incomplete fusion of facial prominences during embryonic development.[1,2] For patients with cleft lip and palate, alveolar cleft reconstruction is a procedure that aims to create a bony bridge that restores dental arch continuity, repair the oronasal fistula, provide support to the structure of the alar base, facilitate subsequent orthodontic treatment, and promote tooth eruption.[3,4] Autologous bone grafting is the gold standard for treating alveolar clefts, with the iliac crest bone being the most widely accepted donor site.[5,6] Other materials such as growth factors, combinations of improved scaffolds and cell treatment/growth factors, biocomposites, and hemostatic agents can be used to regenerate bone and have been the subjects of intensive research.[7–9]

An accurate volumetric analysis method for evaluating the efficiency of different types of alveolar cleft reconstruction is essential because it can help determine which graft material is more effective, confirm favorable times for alveolar bone grafting, and improve surgical techniques. Furthermore, to continue postoperative orthodontic and dental treatments, bone transplant outcomes must be known.[10]

Bone grafting outcomes were previously assessed mainly with the Bergland grading system, which used two-dimensional (2D) radiographs, including panoramic, occlusal, and periapical films, to evaluate the interalveolar bone and relied on linear measurements and subjective evaluations.[7,11] However, the reliability and quality of 2D images for clinical assessment remain questionable and always resulted in an overestimation of the bone formation ratio.[10,12] Currently, the aforementioned problems associated with 2D imaging can be addressed by performing three-dimensional (3D) measurements via computed tomography (CT), which allows visualization of defects in all 3 planes without superposition of the structures.[13]

The development of 3D CT and computer-aided engineering (CAE) has led to new methods of evaluating the efficiency of alveolar cleft reconstruction.[10,14] Surgeons require methods that can precisely analyze the parameters of the newly formed bone, including volume, morphology, and architecture. This study aimed to introduce a novel and precise method of evaluating the outcomes of alveolar cleft reconstruction and grafting procedures using the CAE system.

METHODS

This study was approved by the ethics committee of the Plastic Surgery Hospital Affiliated to Chinese Academy of Medical

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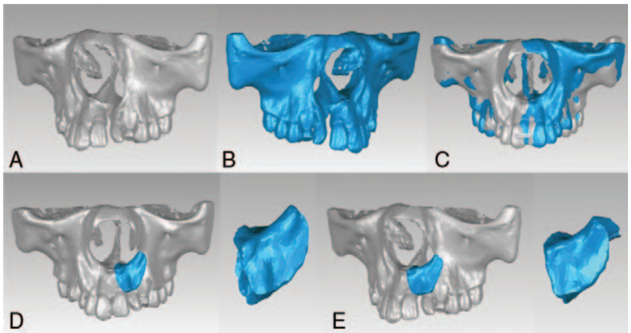

FIGURE 1. Alveolar defect segmented by computer-aided engineering software. The blue regions in (D) and (E) represent the ideal bone defect block. (A) Preoperative maxillary bone. (B) The mirror-reversed maxillary. (C) Accurate registration was performed for the original and mirror-reversed three-dimensional objects. (D) Front view. (E) Isometric view.

Sciences. A 12-year-old boy with a unilateral alveolar cleft who had undergone primary modified Millard cleft lip repair at an early age and who was about to undergo secondary alveolar bone grafting using autologous iliac crest bone was enrolled in this study. The patient did not have any systemic diseases, had good oral hygiene, and had no local pathology in the maxilla that could interfere with surgery.

The patient underwent helical CT (Brilliance 64; Philips, Eindhoven, the Netherlands) preoperatively and 12 months postoperatively. The shape and volume of the alveolar cleft defect were determined using the CAE system, and the standard triangulated language (STL) data from the preoperative CT were analyzed using reverse-engineering software (Geomagic Wrap 2017; Geomagic, Morrisville, NC). To obtain the ideal alveolar bone morphology, the preoperative maxillary bone was mirrored by the software. The original and mirror-reversed 3D objects were precisely adjusted by automatic registration. Then, the bone defect was segmented, and the defect volume (DV) was calculated by the software (Fig. 1).[14] During surgery, the actual bone graft volume (GV) was calculated using the syringe method. The following steps were then performed to evaluate the efficiency of the reconstruction procedure.

Step 1: The preoperative and the postoperative Digital Imaging and Communications in Medicine (DICOM) data from the CT images were processed by the image analysis software (Mimics; Materialise, Leuven, Belgium).
Step 2: The data points were reconstructed as 3D images and saved in the STL format (Fig. 2).
Step 3: The preoperative and the postoperative STL data were processed by Geomagic Wrap 2017 (Geomagic).
Step 4: The first registration was performed for the 3D objects before and after precise grafting, and the 3D objects were superimposed. A second registration was performed based on the middle of the facial bone to obtain the best fit (Fig. 2).
Step 5: A Boolean algebra operation was used to subtract preoperative STL data from postoperative STL data so that bone tissue deformation in the bone grafting region could be obtained.
Step 6: The interference data were deleted around the bone grafting region, and the newly formed actual bone was segmented by identifying the differences between the preoperative and the postoperative images. The 3D image of the newly formed bone was obtained during this step (Fig. 3).
Step 7: The volume of the newly formed bone (NV) was then calculated by the software, and the morphology of the bone block was analyzed intuitively (Fig. 3).

FIGURE 2. Three-dimensional images of a skull with a unilateral alveolar cleft, and registration was performed for the three-dimensional (3D) objects before and after grafting. (A) Front view before and after grafting. (B) Isometric view before and after grafting. (C) The 3D objects were superimposed and a second registration was performed based on the middle of the facial bone. (D) Selected postoperative 3D objects. (E) Selected preoperative 3D objects.

Step 8: The bone survival ratio (BS%) based on the GV recorded during surgery was calculated using the following equation: $BS\% = (NV/GV) \times 100\%$.
Step 9: The ideal DV was calculated by the CAE software using the mirror-reversed technique, and the bone formation ratio (BF%) was calculated using the following equation: $BF\% = (NV/DV) \times 100\%$.

The measurement was repeated 3 times, and the average CAE-derived measurement of the newly formed bone volume was calculated.

RESULTS

The data of the volume measured in triplicate are shown in Table S1 (see Supplemental Digital Content, http://links.lww.com/SCS/A803). The mean volume of the new bone formed postoperatively was 0.387 cm^3. The mean time required for calculating the newly formed bone volume was 23 minutes.

The preoperative alveolar cleft DV was 0.933 cm^3, which was determined using the CAE system. The BF% was 41.4% and the intraoperative GV calculated via the syringe method was 1.0 cm^3; the BS% was 38.7%. The morphology of the newly formed bone block was clear (Fig. 3).

The postoperative 3D image clearly indicated that dental arch continuity was restored, the oronasal fistula was repaired, and the bilateral piriform apertures were almost symmetric because the structure of the alar base was supported (Figs. 2, 3).

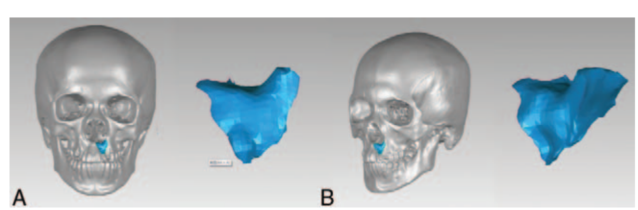

FIGURE 3. Bone tissue deformation in the bone grafting region based on the differences in the preoperative and postoperative standard triangulated language (STL) data. The blue three-dimensional images represent the newly formed bone block (volume = 0.406 cm^3). (A) Front view. (B) Isometric view.

DISCUSSION

In this study, we introduced a novel and precise method of evaluating the outcomes of alveolar cleft reconstruction after grafting procedures using CAE. Alveolar cleft reconstruction is a significant component of the cleft lip surgical treatment protocol. Nowadays, alveolar cleft reconstruction remains controversial with regard to timing, optimal graft materials, surgical techniques, and methods of evaluation,[2] and many studies have intensively investigated these issues.[13,15–17] How to precisely evaluate the outcome of alveolar cleft reconstruction is essential for these studies. Currently, 3D X-ray imaging is the standard method used to verify treatment outcomes of alveolar bone grafts.[10]

Initially, surgeons primarily estimated the outcomes of alveolar cleft reconstruction based on their own experience instead of using objective criteria. This possibly resulted in excessive or deficient graft harvesting.[11] Then, 2D occlusal radiographs were used to show comparable bone growth within the alveolar space.[10,18] The Bergland grading system used a 4-point scale to assess and classify the height of the interalveolar bone graft depending on the coronal level.[11] However, the Bergland scale compared the graft only at the interalveolar level using the normal bone level as a reference.[12] To overcome this disadvantage, the Chelsea scale was developed to evaluate the level of the bone within the cleft compared to the full length of the root surfaces of the teeth next to the cleft and cleft midline at eight sites.[19]

Because of several drawbacks, including distortion in the region around the cleft, lack of volumetric data, an overlap of anatomical structures, problems with magnification, lack of reliable anatomical landmarks, and lack of information regarding the cross-sectional morphology and volume,[12] 2D evaluation was gradually replaced with 3D X-ray diagnostics.[16,20] Recently, 3D volumetric analysis has been widely used to assess the outcomes of bone grafting for patients with clefts.[21] Linderup et al[22] introduced a novel semi-automatic technique for volumetric assessment of alveolar grafts using a third-party software and standardized image acquisition and reconstruction parameters, including anatomical boundaries of the alveolar bone defect. Feng et al[23] made some modifications to this method and elaborated the details of the procedure to make it more precise and practical; they clearly defined more landmarks that were relatively stable in the bone structure and available in a limited field of view. Cranio-caudal and bucco-palatal boundaries are often difficult to discriminate; therefore, Janssen et al[24] attempted to overcome the difficulty of defining the borders of the pre-existing alveolar cleft defect by mirroring the contralateral unaffected side over the alveolar cleft defect to delineate the original bony boundaries of the defect.

However, these existing methods involved indirectly obtaining the newly formed actual bone volume by calculating the postoperative cleft volume around the newly formed bone and subtracting it from the preoperative defect volume. Because of variations in 2 measurements of the anatomical boundaries of the alveolar bone defect, the sources of error increased. Moreover, after complete reossification of the graft, it remained unclear where the former bony boundaries of the defect were situated. This impeded accurate analysis of the residual volume. To our knowledge, no study has reported a direct method of determining the volume of the new postoperatively formed bone.

In our study, the preoperative and 12-months postoperative 3D images were superimposed using the optimal registration feature of the CAE software. The newly formed bone was segmented by identifying the differences between the preoperative and the postoperative 3D images. Using this process, precise anatomical landmarks and well-defined boundaries were unnecessary. This greatly decreased the chances of systemic or manual errors. In addition to the accurate volumetric assessment, this method could be used to obtain visual 3D images of the newly formed bone and could help analyze the morphology of the bone block. This could help determine postoperative orthodontic and dental treatments as well. We preferred performing CT 12 months postoperatively because many studies suggest that the maturation of the cortical structure of the bone graft is complete within 12 months.[10,17,25]

We estimated the defect volumes with the CAE system using mirror-reversed techniques to obtain the ideal alveolar bone morphology. Estimation of the defect size of the alveolar cleft based on the alveolar cleft dimensions of the contralateral side may introduce observer-related variability. Proper alignment of the left-side and right-side structures can be achieved only through image superimposition of a mirrored image, as reported by Janssen et al.[24] Image superimposition of the contralateral side over the defect could also reduce observer dependency when attempting to identify the bucco-palatal dimensions.[24]

An in vivo study revealed a close positive correlation between graft bone volume estimated based on CT data and the actual bone volume measured by syringe during surgery.[26] However, the intraoperative situation is often more complex. Sometimes, the cleft DV can be different from the bone GV. In our study, to evaluate the outcomes of alveolar cleft reconstruction more comprehensively, we used 2 indicators, the BS% and the BF%.

However, there were 2 important limitations associated with this method that need to be resolved. First, helical CT scans involve a high but effective radiation dose, high costs, and long exposure times. Cone beam CT may be suggested as an alternative because of its indisputable advantages, but its resolution is relatively low.[24] Second, after surgery, patients with alveolar clefts still experience skeletal growth and eruption of teeth in the maxillary complex. With an interval of approximately 12 months between preoperative and postoperative scanning, skeletal growth and eruption of teeth might influence the results. In this study, we utilized 2 registrations based on the middle of the facial bone, which is less likely to change with age,[27] to obtain the best fit. Moreover, to limit the measurement bias, the measurement was performed in triplicate and an average was calculated.

CONCLUSIONS

In this study, we described a clinically practical, accurately measurement and time-saving method for evaluating the outcomes of alveolar cleft reconstruction. This method includes volumetric assessment and morphological analysis. Using this method, surgeons and orthodontists can acquire valuable clinical information that can be used to compare various grafting materials and obtain the optimal bone graft volume. This method should be used in future studies to confirm our results.

REFERENCES

1. Moreau JL, Caccamese JF, Coletti DP, et al. Tissue engineering solutions for cleft palates. *J Oral Maxillofac Surg* 2007;65:2503–2511
2. Seifeldin SA. Is alveolar cleft reconstruction still controversial? (Review of literature). *Saudi Dent J* 2016;28:3–11
3. Schnitt DE, Agir H, David DJ. From birth to maturity: a group of patients who have completed their protocol management. Part I. Unilateral cleft lip and palate. *Plast Reconstr Surg* 2004;113:805–817
4. Seike T, Hashimoto I, Matsumoto K, et al. Early postoperative evaluation of secondary bone grafting into the alveolar cleft and its effects on subsequent orthodontic treatment. *J Med Invest* 2012;59:152–165
5. Baqain ZH, Anabtawi M, Karaky AA, et al. Morbidity from anterior iliac crest bone harvesting for secondary alveolar bone grafting: an outcome assessment study. *J Oral Maxillofac Surg* 2009;67:570–575

6. Tan AE, Brogan WF, McComb HK, et al. Secondary alveolar bone grafting–five-year periodontal and radiographic evaluation in 100 consecutive cases. *Cleft Palate Craniofac J* 1996;33:513–518
7. Kamal M, Ziyab AH, Bartella A, et al. Volumetric comparison of autogenous bone and tissue-engineered bone replacement materials in alveolar cleft repair: a systematic review and meta-analysis. *Br J Oral Maxillofac Surg* 2018;56:453–462
8. Khojasteh A, Kheiri L, Motamedian SR, et al. Regenerative medicine in the treatment of alveolar cleft defect: a systematic review of the literature. *J Craniomaxillofac Surg* 2015;43:1608–1613
9. Kolk A, Handschel J, Drescher W, et al. Current trends and future perspectives of bone substitute materials—from space holders to innovative biomaterials. *J Craniomaxillofac Surg* 2012;40:706–718
10. Stasiak M, Wojtaszek-SIomińska A, Racka-Pilszak B. Current methods for secondary alveolar bone grafting assessment in cleft lip and palate patients—a systematic review. *J Craniomaxillofac Surg* 2019;47:578–585
11. Bergland O, Semb G, Abyholm FE. Elimination of the residual alveolar cleft by secondary bone grafting and subsequent orthodontic treatment. *Cleft Palate J* 1986;23:175–205
12. Oberoi S, Chigurupati R, Gill P, et al. Volumetric assessment of secondary alveolar bone grafting using cone beam computed tomography. *Cleft Palate Craniofac J* 2009;46:503–511
13. Chen GC, Sun M, Yin NB, et al. A novel method to calculate the volume of alveolar cleft defect before surgery. *J Craniofac Surg* 2018;29:342–346
14. Du F, Li B, Yin N, et al. Volumetric analysis of alveolar bone defect using three-dimensional-printed models versus computer-aided engineering. *J Craniofac Surg* 2017;28:383–386
15. Saruhan N, Ertas U. Evaluating of platelet-rich fibrin in the treatment of alveolar cleft with iliac bone graft by means of volumetric analysis. *J Craniofac Surg* 2018;29:322–326
16. Liang F, Yen SL, Imahiyerobo T, et al. Three-dimensional cone beam computed tomography volumetric outcomes of rhBMP-2/demineralized bone matrix versus iliac crest bone graft for alveolar cleft reconstruction. *Plast Reconstr Surg* 2017;140:767–774
17. Du F, Wu H, Li H, et al. Bone marrow mononuclear cells combined with beta-tricalcium phosphate granules for alveolar cleft repair: A 12-month clinical study. *Sci Rep* 2017;7:13773
18. Kindelan JD, Nashed RR, Bromige MR. Radiographic assessment of secondary autogenous alveolar bone grafting in cleft lip and palate patients. *Cleft Palate Craniofac J* 1997;34:195–198
19. Witherow H, Cox S, Jones E, et al. A new scale to assess radiographic success of secondary alveolar bone grafts. *Cleft Palate Craniofac J* 2002;39:255–260
20. Feichtinger M, Mossbock R, Karcher H. Assessment of bone resorption after secondary alveolar bone grafting using three-dimensional computed tomography: a three-year study. *Cleft Palate Craniofac J* 2007;44:142–148
21. Blessmann Weber JB, de Macedo Menezes L, Azeredo F, et al. Volumetric assessment of alveolar clefts: a literature review. *J Oral Pathol Med* 2017;46:569–573
22. Linderup BW, Küseler A, Jensen J, et al. A novel semiautomatic technique for volumetric assessment of the alveolar bone defect using cone beam computed tomography. *Cleft Palate Craniofac J* 2015;52:47–55
23. Feng B, Jiang M, Xu X, et al. A new method of volumetric assessment of alveolar bone grafting for cleft patients using cone beam computed tomography. *Oral Surg Oral Med Oral Pathol Oral Radiol* 2017;124:e171–e182
24. Janssen NG, Schreurs R, Bittermann GKP, et al. A novel semi-automatic segmentation protocol for volumetric assessment of alveolar cleft grafting procedures. *J Craniomaxillofac Surg* 2017;45:685–689
25. Sakio R, Sakamoto Y, Ogata H, et al. Effect of platelet-rich plasma on bone grafting of alveolar clefts. *J Craniofac Surg* 2017;28:486–488
26. Shirota T, Kurabayashi H, Ogura H, et al. Analysis of bone volume using computer simulation system for secondary bone graft in alveolar cleft. *Int J Oral Maxillofac Surg* 2010;39:904–908
27. Laowansiri U, Behrents RG, Araujo E, et al. Maxillary growth and maturation during infancy and early childhood. *Angle Orthod* 2013;83:563–571

Virtual Surgical Planning for Mandible Reconstruction With a Double Barrel Fibula Flap and Immediate Implant Placement

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Abstract: This brief clinical report describes our experience with virtual surgical planning in a case of mandibulectomy and mandibular reconstruction with a double barrel vascularized osteofasciocutaneous fibula free flap and immediate implant placement in a case of mandibular ameloblastoma. Fibular segments were positioned to obtain the best result both for masticatory function and for aesthetic facial appearance. Furthermore, in this particular case, as well as being positioned for future masticatory rehabilitation, the implants have served to stabilize the fibula segments in the reconstructive intraoperative phase. A superimposition of programed surgery and 6 months postoperative computed tomography scan was performed and results are presented.

Key Words: Dental implant in fibula flap, double barrel fibula flap, mandibular reconstruction, mandibular rehabilitation, virtual surgical planning

Nowadays, implant rehabilitation is considered to be an integral part of the reconstructive program after resection of tumors of the jaw or for trauma with loss of material that requires reconstruction with osteofascial or composite flaps.[1–7] with regard to the functional aspect, implant-prosthetic rehabilitation must be considered important in patients with maxillary or mandibular reconstruction.[8]

METHODS

Based on the experience of the authors[7,9–12] in virtual planning for mandibular reconstruction and refinement,[11] in this report, we present an unusual case of a 52-year-old woman who was a nonsmoker and in an excellent general state of health but who was affected by a right mandibular ameloblastoma (Fig. 1A) that required right mandibulectomy. Informed consent was obtained from the patient. The work described has been carried out in accordance with the Declaration of Helsinki and was approved by the authors' local institutional review board.

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